Canonical ensemble methods in Thermal-FIST

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1 Overview

The `ThermalModelCanonical` class computes thermodynamics of a hadron resonance gas (HRG) in the canonical ensemble (CE) with exact conservation of baryon number B, electric charge Q, strangeness S, and optionally charm C. The CE partition function reads

$$Z(\mathbf{N}) = \int_{-\pi}^{\pi} \frac{d\phi_B}{2\pi} \int_{-\pi}^{\pi} \frac{d\phi_Q}{2\pi} \int_{-\pi}^{\pi} \frac{d\phi_S}{2\pi}\, e^{\,g(\boldsymbol{\phi})}\,, \tag{1}$$

where $\mathbf{N} = (B, Q, S, \ldots)$ are the conserved charges and

$$g(\boldsymbol{\phi}) = -i\,\mathbf{N}\cdot\boldsymbol{\phi} + \sum_j \sum_{n=1}^{n_{\max}} z_j^n\, e^{\,i\,n\,\mathbf{q}_j\cdot\boldsymbol{\phi}}\,. \tag{2}$$

Mean multiplicities and covariances are obtained from partition function ratios $R(\mathbf{c}) \equiv Z(\mathbf{N} - \mathbf{c})/Z(\mathbf{N})$:

$$\langle N_j \rangle_{\rm CE} = \sum_n n\, z_j^n\; R(n\,\mathbf{q}_j)\,, \tag{3}$$

$$\mathrm{cov}_{\rm CE}(N_j, N_k) = \delta_{jk} \sum_n n^2\, z_j^n\, R(n\,\mathbf{q}_j) + \sum_{n,m} nm\, z_j^n\, z_k^m \Big[R(n\,\mathbf{q}_j + m\,\mathbf{q}_k) - R(n\,\mathbf{q}_j)\, R(m\,\mathbf{q}_k) \Big]\,. \tag{4}$$

Three methods are available, selected via `SetMethod()`:

`CanonicalMethod`	Value	Description
`GaussLegendre`	0	Multi-dimensional numerical quadrature
`SaddlePoint`	1	Per-quantum-number saddle-point solve
`SaddlePointNLO`	2	Analytic LO+NLO from CGF expansion

2 Gauss–Legendre quadrature (`GaussLegendre`)

The multi-dimensional Fourier integral (1) is evaluated by composite 10-point Gauss–Legendre quadrature on $[0,\pi]^d$ (d = number of canonical charges). The interval for each angle is divided into $n_{\max}$ subintervals, with

$$n_{\max} = m \times \max\Big(3,\ \sqrt{B^2 + Q^2 + S^2 + C^2}\Big)\,, \tag{5}$$

where $m \geq 1$ is a user-adjustable multiplier (`SetIntegrationIterationsMultiplier()`).

Baryon analytical integration. When enabled (`m_Banalyt`), the baryon fugacity ϕ_B is integrated analytically using modified Bessel functions, reducing the quadrature dimension by one:

$$\int_0^\pi \frac{d\phi_B}{\pi}\, \cos(B_g\, \phi_B)\, \exp\Big(2|w|\cos(\phi_B - \psi)\Big) = I_{B_g}(2|w|)\,. \tag{6}$$

Saddle-point contour shift. When the baryon analytical path is *not* active (`m_Banalyt = false`), the code shifts the Fourier integration contour through the saddle point before evaluating the GL quadrature. Concretely, `PrepareModelGCE()` solves the GCE constraint equations

$$\sum_j \sum_n n\, q_{a,j}\, \omega_j^{*,n} = N_a \qquad (a \in \text{canonical charges}) \tag{7}$$

for the saddle-point chemical potentials $\boldsymbol{\mu}_{\rm sp}$, and `FillChemicalPotentials()` propagates them to each species. This is equivalent to the contour deformation $\phi_a \to \phi_a + i\mu_{{\rm sp},a}/T$ (Cauchy's theorem for periodic analytic functions) and is mathematically exact.

The effect is to populate the antisymmetric multiplicities $\texttt{Nsy}[\mathbf{q}] = V_c \sum_{j:\mathbf{q}_j=\mathbf{q}} \sinh(\mu_j/T)\, z_j$ in the integrand (Eq. 2). At the saddle point, the condition $\sum_{\mathbf{q}} q_a\, \texttt{Nsy}[\mathbf{q}] = N_a$ ensures that the linear phase term $-N_a\, \phi_a + \sum_{\mathbf{q}} q_a\, \texttt{Nsy}[\mathbf{q}] \sin(q_a\, \phi_a)$ vanishes at $\boldsymbol{\phi} = 0$ to first order. This eliminates the rapid oscillation that occurs at finite chemical potentials (e.g. strangeness-canonical at nonzero μ_B), where the large `Nsy` terms from strange baryons cause severe cancellation in the integrand.

The multiplicative prefactor $e^{-\mathbf{N}\cdot\boldsymbol{\mu}_{\rm sp}/T}$ introduced by the contour shift cancels implicitly: in the partition function ratios $R(\mathbf{c}) = Z(\mathbf{N}-\mathbf{c})/Z(\mathbf{N})$, it leaves a residual factor $e^{\mathbf{c}\cdot\boldsymbol{\mu}_{\rm sp}/T}$ that is exactly compensated by the shifted GCE densities $n_j^{\rm GCE}(\boldsymbol{\mu}_{\rm sp})$ already used in $\langle N_j\rangle_{\rm CE} = R(\mathbf{q}_j)\, n_j^{\rm GCE}$. For entropy, the shift contributes $\ln Z(\mathbf{N}) = -\mathbf{N}\cdot\boldsymbol{\mu}_{\rm sp}/T + \ln \tilde{Z}(\mathbf{N})$; the additional term $-\boldsymbol{\mu}_{\rm sp}\cdot\mathbf{N}/(T\, V_c)$ combines with the shifted thermal contributions to reproduce the correct canonical entropy. No explicit prefactor correction is needed for any thermodynamic observable.

The contour shift is *not* applied in the `m_Banalyt` mode. The Bessel identity used there assumes particle–antiparticle symmetry ($\texttt{Nsx}[\mathbf{q}] = \texttt{Nsx}[-\mathbf{q}]$), which breaks at $\boldsymbol{\mu} \neq 0$. This restriction is benign: `m_Banalyt` is active only when all four charges are canonical, where the integrand at $\boldsymbol{\mu} = 0$ is already symmetric under $\boldsymbol{\phi} \to -\boldsymbol{\phi}$ and has no oscillation problem.

Outputs. The quadrature produces $Z(\mathbf{N}-\mathbf{c})$ for all needed charge vectors $\mathbf{c}$. The ratios $R(\mathbf{c})$ are then fed into Eqs. (3)–(4) to obtain means and covariances.

Limitations. At large correlation volumes ($V_c \gtrsim 5000$ fm^3), catastrophic cancellation in double precision degrades the fluctuation observables: $R(\mathbf{c}+\mathbf{c}')/[R(\mathbf{c})\, R(\mathbf{c}')]$ deviates from unity by only $\mathcal{O}(V_c^{-1})$ but both numerator and denominator are $\sim e^{\mathcal{O}(V_c)}$. The saddle-point contour shift eliminates the oscillation problem at finite $\boldsymbol{\mu}$ but does not resolve the peak-narrowness issue at very large volumes: the integrand peak width scales as $\sim 1/\sqrt{N_{\rm tot}}$, which can be too narrow for the GL quadrature to resolve even with the contour shift. In this regime, the saddle-point methods (Sections 3–4) should be used instead.

3 Saddle-point approximation (`SaddlePoint`)

This method applies the Gaussian saddle-point approximation to each partition function individually, then forms ratios.

3.1 Saddle-point ingredients

Deform the contour $\phi_a = \mu_a/T + i\theta_a$ and define:

$$\omega_j^{*,n} \equiv z_j^n \, \exp\Big(n \, \mathbf{q}_j \cdot \frac{\boldsymbol{\mu}^*}{T}\Big) , \tag{8}$$

where $\boldsymbol{\mu}^*/T$ solves the saddle-point (GCE constraint) equations:

$$\sum_j \sum_n n \, q_{a,j} \, \omega_j^{*,n} = N_a , \qquad a \in \{B, Q, S, \ldots\} . \tag{9}$$

The *susceptibility matrix* is

$$\Sigma_{ab} = \sum_j q_{a,j} \, q_{b,j} \, W_j^{(2)} , \qquad W_j^{(k)} \equiv \sum_n n^k \, \omega_j^{*,n} . \tag{10}$$

3.2 Partition function ratios

For each charge vector $\mathbf{c}$, solve the shifted saddle-point equations for $\boldsymbol{\mu}^*_{\mathbf{c}}$ with charges $\mathbf{N} - \mathbf{c}$ and compute the ratio as

$$\boxed{\ln R(\mathbf{c}) = \frac{V_c}{T} \big[p(\boldsymbol{\mu}^*_{\mathbf{c}}) - p(\boldsymbol{\mu}^*_0) \big] - \Bigg[\sum_a \frac{\mu^*_{\mathbf{c},a}(N_a - c_a)}{T} - \sum_a \frac{\mu^*_{0,a} N_a}{T} \Bigg] - \frac{1}{2} \big[\ln\det\Sigma_{\mathbf{c}} - \ln\det\Sigma_0 \big] ,} \tag{11}$$

where $p(\boldsymbol{\mu}) = \sum_j p_j(T, \mu_j^*)$ is the total GCE pressure and $\Sigma_{\mathbf{c}}$ is evaluated at $\boldsymbol{\mu}^*_{\mathbf{c}}$. Each of the three terms is $\mathcal{O}(1)$, avoiding the catastrophic cancellation that plagues GL at large V_c.

3.3 Algorithm

1. Solve Eq. (9) for $\boldsymbol{\mu}^*_0$ and compute p_0, Σ_0, $\ln\det\Sigma_0$.
2. For each $\mathbf{c} \neq 0$: solve the shifted equations via Broyden's method (warm-started from the previous solution), compute $p_{\mathbf{c}}$, $\Sigma_{\mathbf{c}}$, and evaluate Eq. (11).
3. Feed the $R(\mathbf{c})$ values into Eqs. (3)–(4).

Conservation laws. Mean charges $\sum_j q_{a,j} \langle N_j \rangle_{\mathrm{CE}}$ deviate from N_a by $\mathcal{O}(V_c^{-1})$, since the Gaussian saddle-point for each $R(\mathbf{c})$ is evaluated independently.

4 Saddle-point NLO (`SaddlePointNLO`)

This method computes means and covariances directly from the saddle-point expansion of the cumulant generating function (CGF), bypassing partition function ratios entirely. See the companion note [1] for full derivations.

4.1 Key quantities

At the saddle point $\boldsymbol{\mu}^*/T$ (Eq. 9), define:

- Cluster moments: $W_j^{(k)} = \sum_n n^k \, \omega_j^{*,n} \quad (k = 1, 2, 3, 4)$.
- Susceptibility matrix: $\Sigma_{ab} = \sum_j q_{a,j} \, q_{b,j} \, W_j^{(2)}$.
- Projection: $G_{jk} = \mathbf{q}_j^T \, \Sigma^{-1} \, \mathbf{q}_k \quad [\mathcal{O}(V_c^{-1})]$.
- Response matrix: $D_{jm} = \delta_{jm} - W_m^{(2)} \, G_{jm}$.

4.2 Mean yields (LO + NLO)

$$\boxed{\langle N_l \rangle_{\mathrm{CE}} = W_l^{(1)} - \frac{1}{2} W_l^{(3)} \, G_{ll} + \frac{1}{2} W_l^{(2)} \sum_j W_j^{(3)} \, G_{jj} \, G_{jl} + \mathcal{O}(V_c^{-1}) \, .} \tag{12}$$

The first term is the GCE mean $[\mathcal{O}(V_c)]$. The second and third are the NLO canonical correction $[\mathcal{O}(V_c^0)]$, involving the third cluster moment $W^{(3)}$ (quantum-statistical skewness).

4.3 Covariances (LO + NLO)

$$\mathrm{cov}_{\mathrm{CE}}(N_l, N_m) = \underbrace{\delta_{lm} \, W_l^{(2)} - W_l^{(2)} \, W_m^{(2)} \, G_{lm}}_{\text{LO: } \mathcal{O}(V_c)} \underbrace{- \frac{1}{2} B_{lm} + \frac{1}{2} C_{lm}}_{\text{NLO: } \mathcal{O}(1)} + \mathcal{O}(V_c^{-1}) \, , \tag{13}$$

where the NLO terms involve auxiliary quantities:

$$C_{lm} = \mathrm{Tr}(M_l \, M_m) \, , \quad (M_l)_{ca} = \sum_j W_j^{(3)} \, D_{jl} \, P_{j,c} \, q_{a,j} \, , \quad P_{j,a} = \sum_b (\Sigma^{-1})_{ab} \, q_{b,j} \, , \tag{14}$$

$$B_{lm} = \sum_j G_{jj} \left[W_j^{(4)} \, D_{jl} \, D_{jm} + W_j^{(3)} \left. \frac{\partial^2 \tilde{\alpha}_j}{\partial \alpha_l \partial \alpha_m} \right|_0 \right] , \tag{15}$$

with the second-derivative response

$$\left. \frac{\partial^2 \tilde{\alpha}_j}{\partial \alpha_l \partial \alpha_m} \right|_0 = - \sum_a q_{a,j} \sum_b (\Sigma^{-1})_{ab} \, S_{b,lm} \, , \qquad S_{b,lm} = \sum_k q_{b,k} \, W_k^{(3)} \, D_{kl} \, D_{km} \, . \tag{16}$$

4.4 Algorithm

1. Solve Eq. (9) for $\boldsymbol{\mu}^*/T$ via Broyden's method.
2. Compute $W_j^{(k)}$ $(k = 1, \ldots, 4)$ for all species from the cluster expansion.
3. Build $\Sigma_{ab} = \sum_j q_{a,j} \, q_{b,j} \, W_j^{(2)}$ directly from the $W^{(2)}$ values. Invert to get Σ^{-1}.
4. Compute G_{jj}, $P_{j,a}$, and D_{jl} for all species.
5. Evaluate means from Eq. (12): cost $\mathcal{O}(N_{\mathrm{spec}}^2)$.
6. If fluctuations are requested, evaluate the full covariance matrix from Eqs. (13)–(16): cost $\mathcal{O}(N_{\mathrm{spec}}^3)$.

Conservation laws. The identity $\sum_l q_{a,l} \, D_{jl} = 0$ ensures $\sum_l q_{a,l} \, \langle N_l \rangle_{\mathrm{CE}} = N_a$ exactly, order by order. Similarly, $\mathrm{var}_{\mathrm{CE}}(Q_a) = 0$ is satisfied at LO by $\Sigma - \Sigma \, \Sigma^{-1} \, \Sigma = 0$.

5 Selective canonical treatment

Any subset of charges can be treated canonically while the rest remain grand-canonical. For example, canonical B and S with grand-canonical Q: the saddle-point integral becomes two-dimensional, with a 2×2 susceptibility submatrix. All formulas hold with Σ^{-1} restricted to the canonical subspace and charge vectors projected accordingly.

6 Comparison

	`GaussLegendre`	`SaddlePoint`	`SaddlePointNLO`
Partition functions	GL quadrature	per-$\mathbf{c}$ SP solve	not computed
Mean yields	from $R(\mathbf{c})$	from $R(\mathbf{c})$	direct (Eq. 12)
Fluctuations	from $R(\mathbf{c}+\mathbf{c}')$	from $R(\mathbf{c}+\mathbf{c}')$	analytic (Eq. 13)
Exact conservation	up to quad. error	broken at $\mathcal{O}(V_c^{-1})$	exact by construction
Finite-$\boldsymbol{\mu}$ stability	contour shift	native	native
Large-V_c stability	degrades	stable	stable
Small-V_c validity	exact	breaks down	breaks down
Timing (full HRG, $\sim$500 species, $V_c = 5000$ fm^3):			
Means only	$\sim$ 10 ms	$\sim$ 20 ms	$\sim$ 20 ms
With fluctuations	$\sim$ 18–140 s	$\sim$ 1.5 s	$\sim$ 1 s

When to use which.

- **GaussLegendre**: Small to moderate volumes ($V_c \lesssim$ 100–1000 fm^3) or when the saddle-point approximation is not reliable. The saddle-point contour shift enables use at finite $\boldsymbol{\mu}$ for partial canonical ensembles. Increase the multiplier m for better precision.
- **SaddlePoint**: Large volumes where GL suffers from cancellation. Best when exact partition function ratios $R(\mathbf{c})$ are needed (e.g. for higher-order cumulants beyond covariances).
- **SaddlePointNLO**: Large volumes when exact conservation of mean charges matters (e.g. thermal fits). Fastest for fluctuations. Does not compute individual $R(\mathbf{c})$ values.

References

[1] V. Vovchenko, "Saddle-point approximation for the canonical partition function: cumulant generating function, means, correlations, and fluctuations", companion note (2025).

[2] V. Vovchenko, B. Dönigus and H. Stoecker, Phys. Rev. C **100**, 054906 (2019), arXiv:1906.03145 [hep-ph].